

Milestone 2 — Backend CRUD Implementation (Node.js, Express, Handlebars, MongoDB)

Goal

Implement a full-stack airline reservation system with persistent data storage using Node.js, Express, Handlebars, and MongoDB.

The system should support user accounts, flight management, flight booking, and reservation management. Data should no longer be stored as static or dummy data and must be retrieved from MongoDB.

1. User Management

Passenger Features

Users should be able to:

- Register an account
- Login and logout
- View profile information
- Update profile information

Admin Features

Administrators should be able to:

- Manage flights
 - View reservations
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2. Flight Management

Administrators should be able to perform CRUD operations for flights.

Flight Information

Each flight should contain the following information:

- Flight Number
- Airline
- Origin
- Destination
- Departure Date and Time
- Arrival Date and Time
- Available Seats
- Ticket Price

Required Features

Administrators can:

- Create flights
- View flights
- Update flights
- Delete flights

3. Flight Search

Passengers should be able to search available flights using:

- Origin
- Destination
- Departure Date

Search Results

Search results should display:

- Flight Number
- Airline
- Origin
- Destination
- Departure Date and Time
- Arrival Date and Time
- Available Seats
- Ticket Price

Each result should provide:

- View Details
- Book Flight

4. Flight Booking

Passengers should be able to book a flight.

Booking Information

The booking process should collect:

- Full Name
- Email Address
- Passport Number
- Selected Seat

Booking Workflow

Search Flight



View Flight Details



Enter Passenger Information



Select Seat



Confirm Booking

5. Reservation Management

Passengers should be able to manage their reservations.

Required Features

Passengers can:

- View reservations
- Update seat selection
- Cancel reservations

Reservation Information

Each reservation should display:

- Reservation Number
 - Flight Number
 - Passenger Name
 - Seat Number
 - Booking Status
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6. Business Rules

The following rules must be enforced:

Flight Availability

Bookings cannot exceed the number of available seats.

Seat Availability

A seat may only be assigned to one passenger.

Required Information

The following fields are mandatory:

- Full Name
- Email Address
- Passport Number

Valid Flight

Passengers cannot book flights that have no available seats.

7. AJAX Requirements

Students must use AJAX for at least the following features:

Flight Search

Search results should update without requiring a page refresh.

Seat Selection

Seat availability should update dynamically.

Reservation Cancellation

Reservations should be cancelled without requiring a page refresh.

8. Validation

Client-Side Validation

Validate:

- Required fields
- Email format
- Empty inputs

Server-Side Validation

Validate:

- Flight availability
- Seat availability
- Required passenger information

All validations must also be performed on the server even if they are implemented on the client.

9. Views

Passenger Pages

- Login
- Registration
- Profile
- Flight Search
- Flight Details
- Booking Form
- My Reservations

Admin Pages

- Flight Management
- Reservation List

10. Deliverables

Backend

- Express Routes
- Controllers
- Models

Database

MongoDB collections for:

- Users
- Flights
- Reservations

Frontend

- Handlebars templates
 - Bootstrap-based user interface
 - AJAX implementation for required features
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Milestone 2 Deliverables Summary

By the end of Milestone 2, the system should allow:

1. Users to register, login, and manage their profile.
2. Administrators to manage flights.
3. Passengers to search and book flights.
4. Passengers to view and cancel reservations.
5. Data to be stored and retrieved from MongoDB.
6. Selected features to use AJAX without requiring page refreshes.
7. Business rules and validation to be enforced on both the client and server sides.